from MSG-20 Q19 (SLD-001 sh 2)

Q19
VCB 630 A
CT 200/1
5P20 + 0.2S

VT 20/√3 / 0.1/√3 kV
0.2 / 3P

R-B1
50/51 · 50N/51N · 67N
27/59 · 81U/O · 59N
ROCOF / VS (LoM)
25 (GFM mode only)
intertrip ← R-INC1

Wh 0.2S revenue

SA 24 kV

3×1×240 Al 12/20 kV

TR-B1
5.5 MVA 20/0.69 kV
Dyn11 · uk 8 % (A)

Q-B1
ACB 5 000 A

PCS-1
5.5 MVA, 4-quadrant
grid-following (+GFM option)

DC 1 500 V

BESS-1
5 MW / 10 MWh LFP
4 × 2.5 MWh containers (A)

from MSG-20 Q29 (SLD-001 sh 2)

Q29
VCB 630 A
CT 200/1
5P20 + 0.2S

VT 20/√3 / 0.1/√3 kV
0.2 / 3P

R-B2
50/51 · 50N/51N · 67N
27/59 · 81U/O · 59N
ROCOF / VS (LoM)
25 (GFM mode only)
intertrip ← R-INC2

Wh 0.2S revenue

SA 24 kV

3×1×240 Al 12/20 kV

TR-B2
5.5 MVA 20/0.69 kV
Dyn11 · uk 8 % (A)

Q-B2
ACB 5 000 A

PCS-2
5.5 MVA, 4-quadrant
grid-following (+GFM option)

DC 1 500 V

BESS-2
5 MW / 10 MWh LFP
4 × 2.5 MWh containers (A)

EMS / PPC priorities
1. Import ≤ 40 MW incl. recharge
2. SOC reserve before peaks
3. FCR down headroom
4. Markets (FCR, mFRR)

Fallback: IT power capping
→ R-B1, R-B2, PCS-1/2

BESS SCHEDULE

Item	Rating (Rev P2)	Note
Bays Q19 / Q29	VCB 630 A, CT 200/1 5P20 + 0.2S	unit 159 A at 20 kV
TR-B1 / TR-B2	5.5 MVA 20/0.69 kV Dyn	delta at 20 kV
PCS-1 / PCS-2	5.5 MVA each, 11 MVA total	p.f. ≥ 0.95 at PCC
BESS-1 / BESS-2	5 MW / 10 MWh LFP each	10 MW / 20 MWh
I″k contribution	≤ 1.2 × In (PCS limit), 13.0 → 13.3 kA	CALC-004

NOTES
1. Not counted as Tier III / EN 50600 Class 3 capacity; firm IT gain +0.84 MW (peak shaving +2.4 MW).
2. No island operation in the base design: 20 kV relays do not clear an inverter-fed island fault (381 A, R-FDR ≈ 60 s).
3. Containers ≥ 10 m (A) from halls, transformers, fuel; UL 9540A, NFPA 855, IEC 62933-5-2.
4. Harmonic study (IEC/TR 61000-3-6) open.

P2	2026-10-25	CALC-002/3/4 ratings, IEC symbols, ≥2.5 mm text
P1	2026-10-24	First concept issue

Nordic Hyperscale DC Campus (fictional) · 30 MW IT · SE3
Portfolio D. Wu · public standards and [ASSUMED] data only

BESS plant — 10 MW / 20 MWh
2 units, MSG-20 bays Q19 / Q29, protection and EMS
CONCEPT · NOT FOR CONSTRUCTION · scale NTS · A3

Drawing no.	Sheet	Rev
DWU-DC-E-SLD-004	**1 / 1**	**P2**